Validation

We have worked with the system of two differential equations for the rainbowfish and gouramis.

$$\begin{cases} \dfrac{dP}{dt} = 0.7\,P - 0.007\,P^2 - \alpha\,PG, & \text{where} \quad \alpha = 0.04, \\ \dfrac{dG}{dt} = -0.25\,G + \beta\,PG, & \text{where} \quad \beta = 0.008. \end{cases}$$

You have learned in §3.1 how to approximate solutions numerically with Euler's method. In this section, we have used analytical methods to characterize the solutions beforehand. First the equilibrium points (P_0, G_0) were calculated:

$$(0,0),\ (100,0) \text{ and } \left(\frac{1}{4\beta}, \frac{0.7}{\alpha}(1 - \frac{1}{400\beta})\right).$$

Then the differential equations were linearised around the three equilibrium points:

$$\text{For} \quad \vec{M}(t) = \begin{bmatrix} P(t) - P_0 \\ G(t) - G_0 \end{bmatrix},$$

$$\frac{d\vec{M}}{dt} \approx \begin{bmatrix} 0.7 - 0.014P_0 - \alpha G_0 & \alpha P_0 \\ \beta G_0 & -0.25 + \beta P_0 \end{bmatrix} \vec{M}(t).$$

By calculating the eigenvalues of the three matrices, the types of the three equilibrium points were determined. In the following exercises, you are going to review whether these analytical calculations were usefull.

Exercise A (1 point possible)

Question

Answer » Reflection » **Results**

Correct Answer

No, probably only the stable equilibrium point, because

Because there is only one stable equilibrium point, and in this case the solutions are bounded, all solutions will end up in that stable equilibrium point. The unstable equilibrium points can only be found when you would accidentally start exactly in them, and even then, even the tiniest numerical error, will make that the solution leaves the equilibrium point.

Your initial answer: No, probably only the stable equilibrium point, because (Option 2)

"it only converges to the solution point"

Your final answer: No, probably only the stable equilibrium point, because (Option 2)

"it only converges to the solution point"

See how your classmates answered below

Class Breakdown

This is a look at how your classmates answered the question during the initial and final rounds.

Answer Options

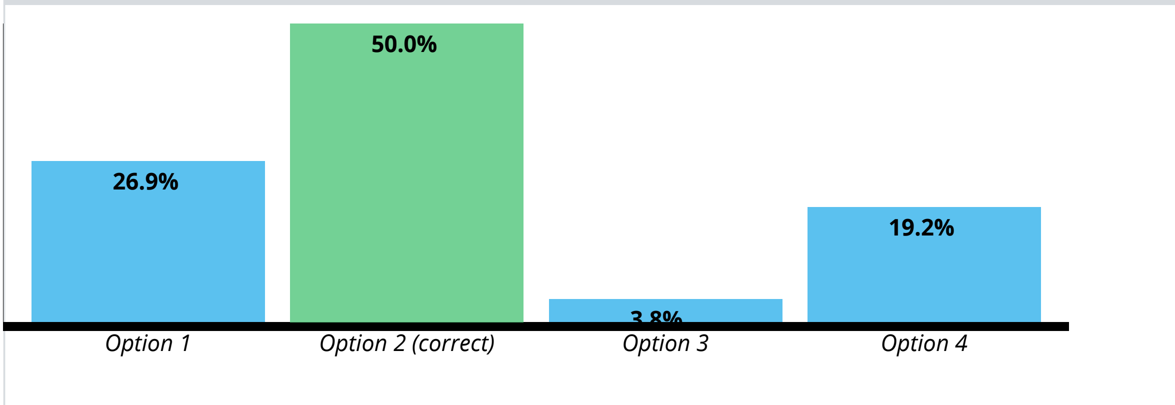
Option 1
Yes, all three of them, because

Option 2 (correct)

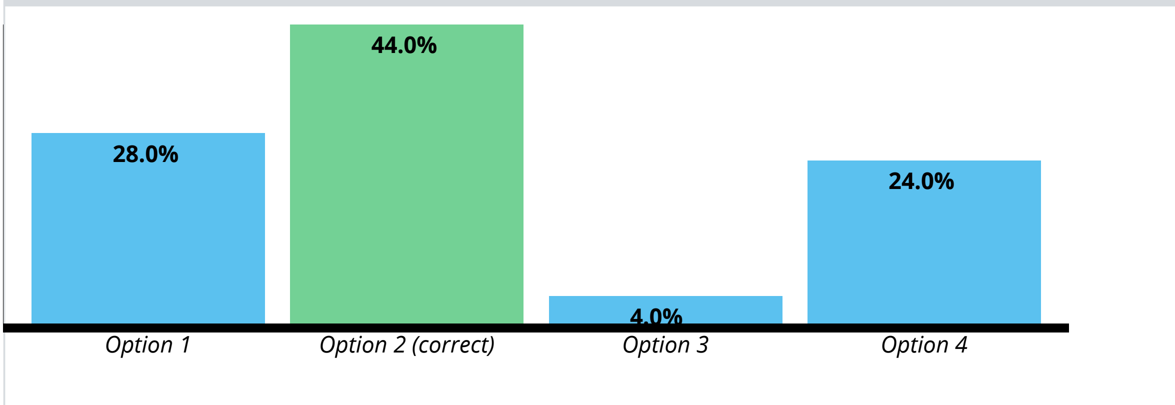
Option 3
No, probably only the instable equilibrium points, because

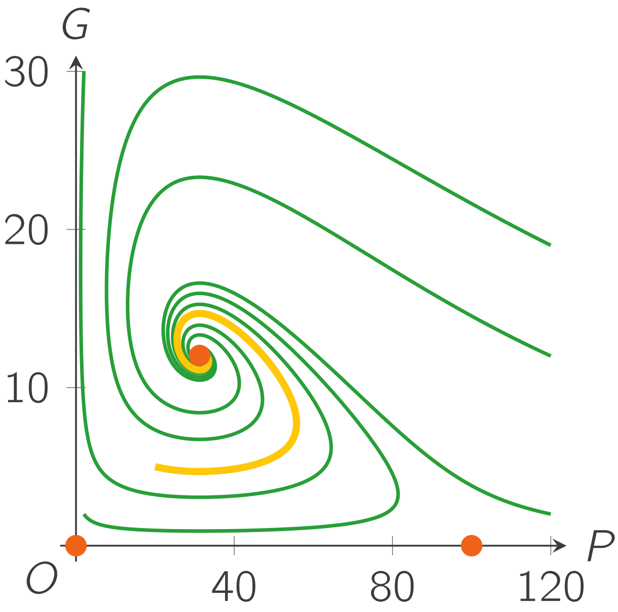
Option 4
No, probably none of the equilibrium points, because

Initial Answer Selection



Final Answer Selection





For the next exercise, draw for yourself the phase lines for the populations in separate aquariums:

- a phase line for P when $\dfrac{dP}{dt} = 0.7\,P - 0.007\,P^2$, and
- a phase line for G when $\dfrac{dG}{dt} = -0.25\,G$.

Exercise B

0.0/2.0 points (graded)

Consider the two saddle points $(0, 0)$ and $(100, 0)$.

Which of the following statements is true?

- [x] When you add arrows, the P-axis in the phase plane is the same as the phase line for P when $\alpha = 0$. ✔
- [x] When you add arrows, the G-axis in the phase plane is the same as the phase line for G when $\beta = 0$. ✔
- [ ] Without rainbowfish in the aquarium, the number of gouramis will decrease, so when there are rainbowfish present in the aquarium, the number of gouramis will decrease as well.
- [ ] The rainbowfish without any gouramis have a stable equilibrium at $P = 100$, so in

- [ ] Without calculating the eigenvalues, you know that equilibrium point $(100, 0)$ can never be a spiral or circle point. ✔

- [ ] When there are approximately 100 rainbowfish and only a few gouramis, the number of gouramis will always first decrease, and the number of rainbowfish will always first go nearer to 100.

- [ ] With different interaction constants α and β, the positions of one or both of the saddle points change.

✖

Submit You have used 3 of 3 attempts

Answers are displayed within the problem

Exercise C

2/2 points (graded)

Consider the equilibrium point $(31.25, 12.03)$.

Linearisation has given you the Jacobian matrix, and its eigenvalues $\lambda \approx -0.11 \pm 0.33i$.

Which of the following statements can you conclude to be true?

- [ ] Complex numbers do not exist in the real world, so the solutions of the linearized system are not valid.

- [x] $-0.11 < 0$, so the equilibrium point is stable. ✔

- [ ] $-0.11 < 0$, so the equilibrium point is unstable.

- [x] For $t \to \infty$ all solutions that are "near" the equilibrium point will end up in it. ✔

- [ ] The solutions near the equilibrium will oscillate, because $-0.11 \neq 0$.

- [x] The solutions near the equilibrium will oscillate, because $0.33 \neq 0$. ✔

- [ ] The trajectories around the equilibrium are too angular and asymmetric to represent an elliptic spiral, so the equilibrium is not a spiral point.

- [x] With different interaction constants α and β, the position of the spiral point will be different. ✔

- [x] With different interaction constants α and β, probably the μ and ω will change, and with very different parameters, even the type of the equilibrium point might change. ✔

✔

Answer

Correct: Well done!

Submit You have used 3 of 3 attempts

Answers are displayed within the problem

In the validation of §3.1 we gave you formulas that approximately describe the behaviour of $P(t)$ and $G(t)$. There you tried to estimate its constants from the simulated results. In the next exercise you will see which constants you can estimate with the analytical results of this section.

Before you start, you should know one more thing: a sine and a cosine that have the same period can be combined to one sine with a phase shift: so factors $e^{at}\cos(bt)$ and $e^{at}\sin(bt)$ can be combined to one factor $e^{at}\sin(bt-\gamma)$. (Again, we will not go into the details of this.)

Exercise D

2.0/2.0 points (graded)

The formulas that approximately describe the evolution of the populations (with $P(0) = 20$ and $G(0) = 5$) are the following ones:

$$
\begin{aligned}
P(t) &\approx A + B\,e^{-\mu t}\,\sin(\omega(t-\phi)),\\
G(t) &\approx C + D\,e^{-\mu t}\,\sin(\omega(t-\psi)).
\end{aligned}
$$

Linearisation around equilibrium point $(31.25, 12.03)$ has given you the Jacobian matrix

- [x] A and C ✔
- [ ] B and D
- [x] μ ✔
- [x] ω ✔
- [x] Cycle length T ✔
- [x] Frequency of the cyclic movement f ✔
- [ ] ϕ and ψ
- [ ] Phase lag $\psi - \phi$

✔

Answer

Correct: Well done! You see that a lot can be learned from the analytical calculations.

Submit You have used 3 of 3 attempts

Answers are displayed within the problem

Next, a page for questions or remarks you might have about this section.